Year 8
Mathematics

Unit 3: Working with Powers

3.1 Simplifying Expressions

Simplifying Expressions

STARTER · 5 MIN

Date: _____ **Title: Simplifying Expressions**

LEARNING OBJECTIVES

- To be able to simplify expressions involving powers and brackets

STARTER *Answer in your book. No calculators.*

1. Simplify $3x + x$
2. Simplify $4 \times 5x$
3. Expand $3(x + 1)$
4. Simplify $7a - 5a$

Starter — Answers

STARTER

Tick or correct in green pen

1. $4x$

2. $20x$

3. $3x + 3$

4. $2a$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can collect like terms correctly	☐
SILVER	I can correctly expand single brackets using multiplication	☐
GOLD	I can form and simplify expressions for areas and perimeters of shapes	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Simplifying Expressions

INTRODUCTION · 3 MIN

To simplify, collect like terms so the expression has as few terms as possible.

Like terms

Same letters and powers.

$3x^2$ and $5x^2$ are like terms

Collect

Add or subtract like terms.

$3x + 5 + 4x - 2 = 7x + 3$

Expand

Multiply every term in the bracket.

$-2(a - 4) = -2a + 8$

Expand and simplify

Expand, then collect.

$2(x + 3) + 3(x + 1) = 5x + 9$

Examples 1 and 2 — Collecting Like Terms

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Like terms: same letters and powers x^2 and x are unlike

Example 1

Simplify

$3x + 5 + 4x - 2$

Example 2

Simplify

$12x^2 - 7x^2 + 3x$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Group the x terms and the numbers

$$3x + 4x + 5 - 2 \quad = 7x + 3$$

Tip: keep each sign with its term.

Example 2

x^2 and x are unlike terms

$$12x^2 - 7x^2 + 3x \quad = 5x^2 + 3x$$

Tip: only collect matching powers.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify*

Hint Like terms: same letters and powers x^2 and x are unlike

1. $5a + 3a - 2a$

2. $4y + 7 - y - 3$

3. $6x^2 + 2x^2$

4. $3x^2 + 2 + x^2 + 5x$

5. $8p - 3q + p + 5q$

★ $2a^2 + 3a - a^2 - 5a$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $6a$

2 $3y + 4$

3 $8x^2$

4 $4x^2 + 5x + 2$

5 $9p + 2q$

★ $a^2 - 2a$

Examples 3 and 4 — Expanding and Simplifying

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Multiply every term − × − = +

Example 3

Expand

$-2(a - 4)$

Example 4

Expand and simplify

$2(x + 3) + 3(x + 1)$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiply both terms by −2

$-2 \times a = -2a$

$-2 \times -4 = +8$

$\mathbf{-2(a - 4) = -2a + 8}$

Tip: − × − = +.

Example 4

Expand both, then collect

$\mathbf{2x + 6 + 3x + 3 = 5x + 9}$

Tip: 6 + 3 = 9.

Task 2

PRACTICE · 9 MIN

SILVER *Expand and simplify*

Hint Multiply every term − × − = +

1. $3(x + 5)$

2. $4(2y - 3)$

3. $-3(m + 2)$

4. $3(2m + 4) + 2(m - 3)$

5. $5(2x + 2) - 3(x - 4)$

★ $6(2x - 3) - (4x - 7)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $3x + 15$

2 $8y - 12$

3 $-3m - 6$

4 $8m + 6$

5 $7x + 22$

★ $8x - 11$

Examples 5 and 6 — Areas and Perimeters

EXAMPLES · 5 MIN

GOLD

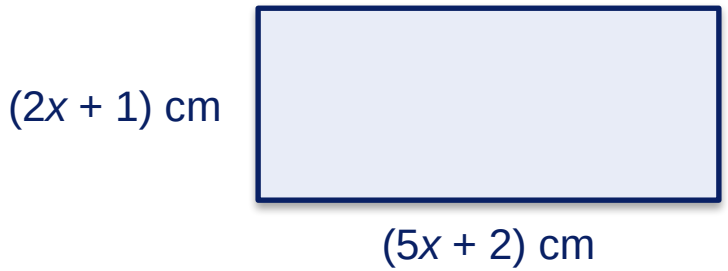
Exam-style: show all your working

Hint Perimeter: add all sides Area: length × width

Example 5

(2 marks)

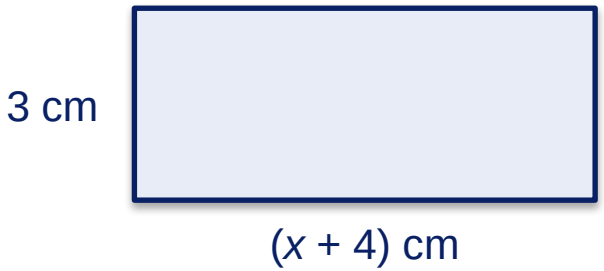
Write and simplify an expression for the perimeter.



Example 6

(2 marks)

Write and simplify an expression for the area.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Add all four sides

$$P = 2(5x + 2) + 2(2x + 1)$$
$$= 10x + 4 + 4x + 2$$
$$= \mathbf{14x + 6}$$

Tip: opposite sides are equal.

Example 6

Area = length × width

$$A = 3(x + 4)$$
$$= \mathbf{3x + 12}$$

Tip: the units are cm^2.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: areas and perimeters*

Hint Perimeter: add all sides Area: length × width

1. Perimeter of a square of side $(x + 3)$

2. Area of a rectangle 4 by $(2x - 1)$

3. Perimeter of a rectangle $(3x + 1)$ by x

4. Area of a rectangle x by $(x + 5)$

5. Triangle sides $2x$, $2x + 1$ and $3x - 4$. Perimeter?

★ Perimeter 34 when $x = 2$: is $14x + 6$ right?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $4x + 12$
2. $8x - 4$
3. $8x + 2$
4. $x^2 + 5x$
5. $7x - 3$

★ Yes: 28 + 6 = 34

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $5a + 3a - 2a$

SILVER Simplify $2(x + 3) + 3(x + 1)$

GOLD Expand $3(x + 5)$ and $-2(a - 4)$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **$6a$**

SILVER **$5x + 9$**

GOLD **$3x + 15$ and $-2a + 8$**

Exam Practice — Simplifying

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Like terms: same letters and powers x^2 and x are unlike

Question 1

(1 mark)

Simplify $12x^2 - 7x^2$

.....

Question 2

(2 marks)

Simplify $3x^2 + 2 + x^2 + 5x$

.....

Exam Practice — Expand and Simplify

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Multiply every term − × − = +

Question 3

(2 marks)

Expand and simplify

$3(2m + 4) + 2(m - 3)$

.....

Question 4

(2 marks)

Expand and simplify

$5(2x + 2) - 3(x - 4)$

.....

Exam Practice — Perimeter and Area

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Perimeter: add all sides
Area: length × width

Question 5

(2 marks)

A square has sides of $(3x + 2)$ cm.

Write a simplified expression for the perimeter, P.

.....

Question 6

(3 marks)

A rectangle is $(2x + 5)$ cm long and 4 cm wide.

(a) Write an expression for its area. (1)

.....

(b) Find the area when $x = 3$. (2)

..... cm^2

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$5x^2$ B1

Tip: 12 − 7 = 5.

Question 2

$4x^2 + 5x + 2$ B2

B1 for two terms correct

Tip: x^2, x and numbers are unlike.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$6m + 12 + 2m - 6$ — M1

$8m + 6$ — A1

Tip: expand both brackets first.

Question 4

$10x + 10 - 3x + 12$ — M1

$7x + 22$ — A1

Tip: $-3 \times -4 = +12$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$P = 4(3x + 2)$ — M1

$= 12x + 8$ — A1

Tip: a square has 4 equal sides.

Question 6

(a) $8x + 20$ — B1

(b) $8 \times 3 + 20 = 44$ cm² — M1 A1

Tip: substitute into your expression.